

DeepSeek CLI Harness 实测对比

Same-prompt comparison of Pi, Codex, and DSH running DeepSeek V4 Flash/Pro at high/max effort, including adjusted execution time, cache use, tool reliability, implementation quality, and preview-CLI recommendations.

Technical Summary

同一 canonical prompt、12 个单次样本下，DSH 的输出质量并不差，但执行可靠性明显落后：最佳 DSH 输出为 Flash/max（88/100），接近 Pi Flash/max（89）；然而 DSH 4/4 都需要两次人工“继续”，Pi/Codex 8/8 均 one-shot。扣除可识别的断线与重试等待后，DSH 仍需 40.3–72.1 分钟，Pi 为 14.1–34.6 分钟，Codex 为 10.3–31.7 分钟。由于模型档位一致，这组差异主要指向 harness 的任务生命周期、工具契约与恢复策略。

Source: Joined performance and quality evidence

Tables: analysis/analyze-runs.mjs, analysis/quality-assessment.md, analysis/chart-map.md

净执行时间：去掉明确等待后，DSH 仍慢约 2–4×

最可比的 Pro/high 组合中，Codex 为 10:19、Pi 为 17:30、DSH 为 40:19；DSH 墙钟时间原为 61:36，本报告按要求扣除了 21:17 的显式 stream-idle timeout、retry backoff 和错误轮次间隔。DSH 的 turn 记录存在缺失结束事件，因此 adjusted time 是保守近似，不应解读为精确 CPU 活跃时间。

Source: Normalized benchmark session analysis

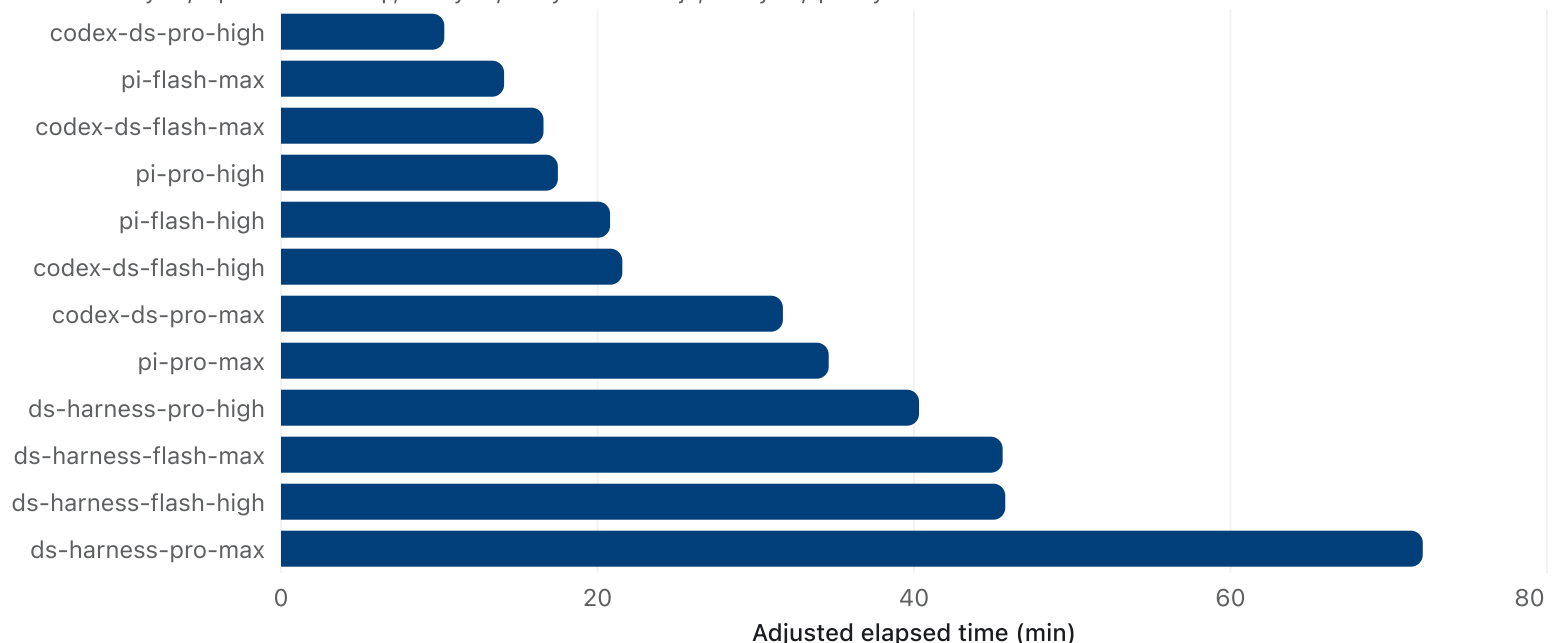
Tables: runs/20260813193424/*/.benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/.benchmark-runtime/{pi,codex}/sessions/**/*/.jsonl, runs/dsh/*/session.jsonl

Adjusted completion time

DSH needs 40.3–72.1 active minutes after removing explicit waits; all four DSH artifacts still require two manual continuations.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



Harness 可靠性：问题不在“报错次数多”，而在错误是否终止任务

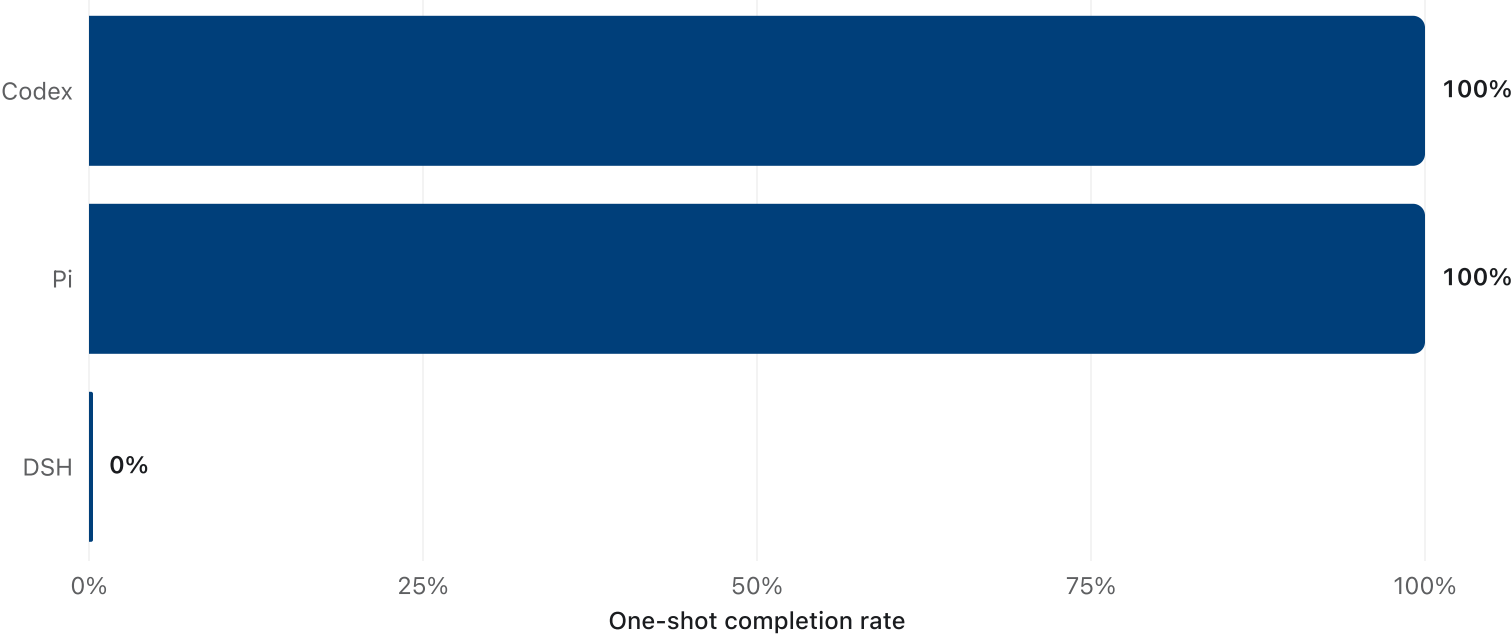
DSH 原始工具错误率只有 9/178 (5.1%)，低于 Pi 的 13/94 (13.8%) 和 Codex 的 17/93 (18.3%)；但 DSH one-shot 为 0/4，另外两者为 8/8。最强反例是 DSH Pro/high：18 次工具调用 **0 次显式工具错误**，仍因流超时与续跑状态机无法一次完成。真正的 harness 差距是错误分类、自动恢复和 turn 生命周期，而不是单一错误率。

Source: Normalized benchmark session analysis
Tables: runs/20260813193424/*/.benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/.benchmark-runtime/{pi,codex}/sessions/**/*.*.jsonl, runs/dsh/*/session.jsonl

One-shot completion rate by harness

Pi and Codex finish 8/8 combined without intervention; DSH finishes 0/4 without two manual continuations.

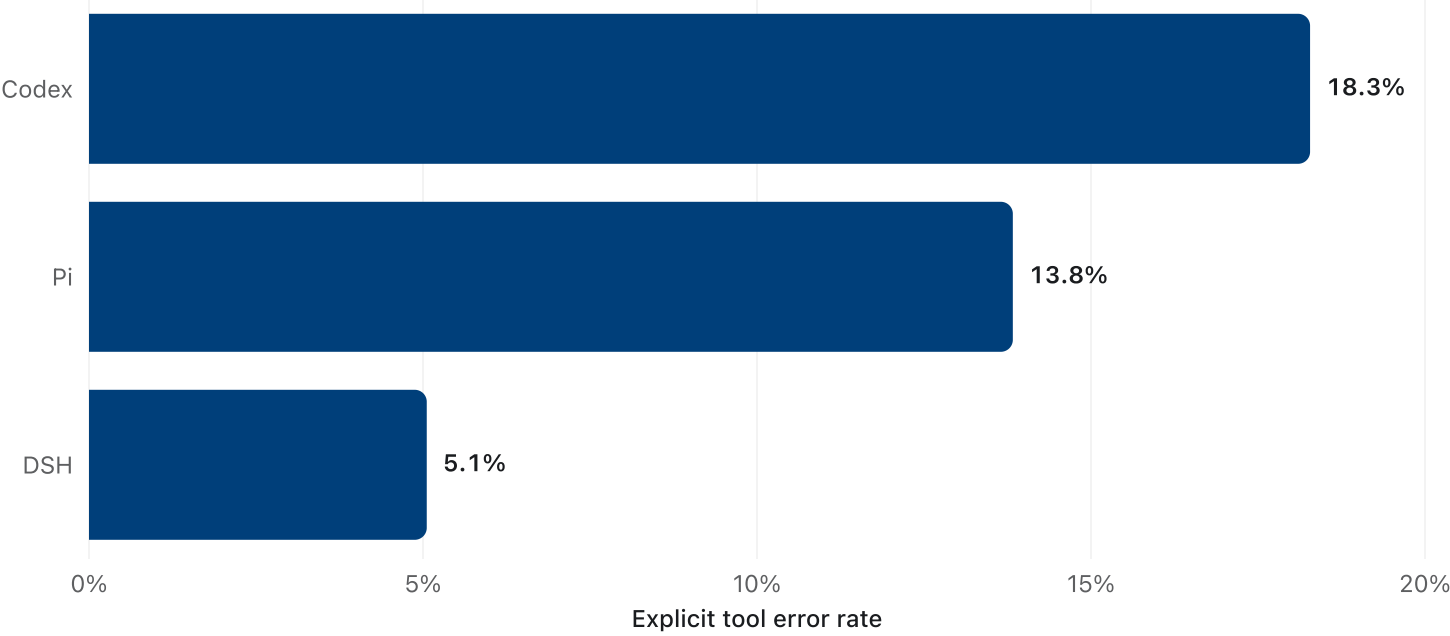
Source: Harness-level reliability query
Tables: analysis/harness-summary.sql, analysis/report-source.sql



Observed tool-result failure rate

DSH has the lowest raw incidence, proving that failure severity and recovery—not error count alone—drive its reliability gap.

Source: Harness-level reliability query
Tables: analysis/harness-summary.sql, analysis/report-source.sql



DSH preview CLI：面向一次完成的友好优化建议

P0 — 先修任务连续性与恢复语义

把 300 秒 idle timeout 升级为 progress-aware watchdog：token、reasoning、tool heartbeat 任一有进展就续租；连接中断后在同一 turn 内自动 resume，并以幂等 step/turn ID 写入原子 completion marker。四组 DSH 共记录 15 次 LLM retry、20 次 timeout occurrence 和 8 次人工“继续”，这是当前最大损耗。

P0 — 基于模型能力协商工具，而不是让模型试错

DeepSeek V4 Flash/Pro 都是纯文本模型。DSH 仍向其暴露 read_image，3/4 case 调用后得到永久 capability error。启动时应根据 model capability registry 隐藏不兼容工具；视觉任务改为外部 render_validate oracle，返回 console errors、DOM/viewport bounds、状态机断言、像素占用与结构化 JSON。需要语义审图时，可选独立 vision evaluator，但不要伪装成主模型能力。

P1 — 让工具契约自带可恢复性

5 次 stale-file edit 应由 wrapper 携带 revision token，并在冲突时自动 re-read + 单次重放；regex 不支持 lookbehind 时应返回可直接执行的 --pcre2 修复提示。统一错误 taxonomy：retryable、permanent、capability_mismatch、permission_denied，由 harness 决定自动恢复预算。

P1 — 预先固化权限与 headless benchmark 模式

3/4 DSH case 在中途切到 danger-full-access。提供显式 --sandbox、--approval、--non-interactive preflight，并在首个模型调用前输出可用工具/路径/浏览器能力，避免运行中改变上下文。

P1 — 对齐 Pi/Codex 的可观测性

提供稳定 --json-summary：resolved CLI/model version、effort、adjusted/wall time、TTFT、token/cache、tool errors by class、retries、timeout、exit reason、artifact paths。回归门槛应优先看 one-shot completion 与 automatic recovery，再看 raw error rate。

Source: Normalized benchmark session analysis

Tables: runs/20260813193424/*/benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/benchmark-runtime/{pi,codex}/sessions/**/*.*.jsonl, runs/dsh/*/session.jsonl

缓存：高命中率无法抵消重复上下文重放

12 个 case 的 cache hit 都高于 96.8%。真正拉开资源消耗的是模型调用次数与总输入：DSH Pro/max 达到 11.03M 输入、75 次模型调用和 72.1 分钟净执行时间；同档 Codex 为 3.53M/23 次，Pi 为 2.91M/20 次。缓存 token 不是零成本，高 cache hit 也不代表高效率。

Source: Normalized benchmark session analysis

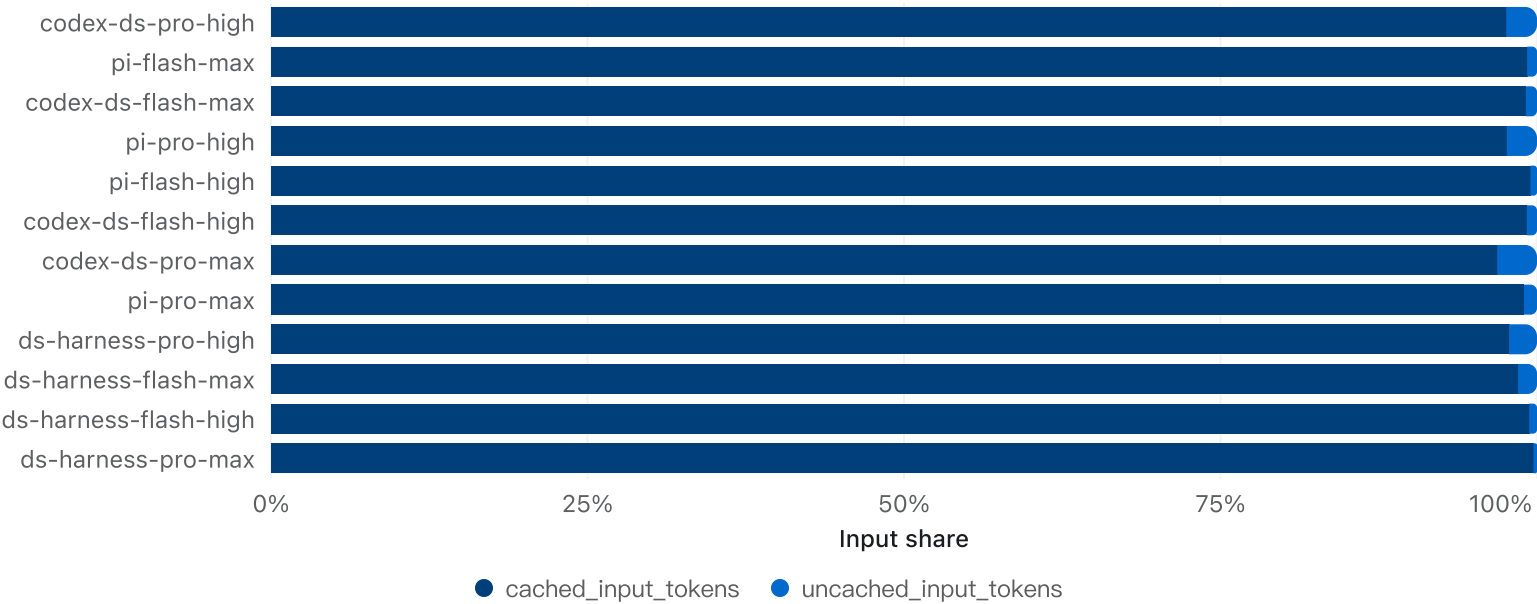
Tables: runs/20260813193424/*/benchmark-runtime/pi/sessions/*.jsonl, runs/20260813195515/*/benchmark-runtime/{pi,codex}/sessions/**/*.*.jsonl, runs/dsh/*/session.jsonl

Input token cache composition

Cache hit is high across all 12 runs, but DSH Pro/max still replays 11.0M input tokens across 75 model calls.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



实现质量：DSH 能做出好结果，但缺少可靠验收闭环

DSH Flash/max (88) 和 Pro/high (87) 已进入第二梯队，说明模型生成能力不是主要瓶颈。DSH Flash/high 的 assembled shutter 脱离壳体，Pro/max 则严重缩小主体并让标签/右侧标题拥挤。三组 DSH 虽生成截图，却因纯文本模型无法直接读取；最终评分来自外部浏览器 QA。

Source: Browser and implementation quality assessment

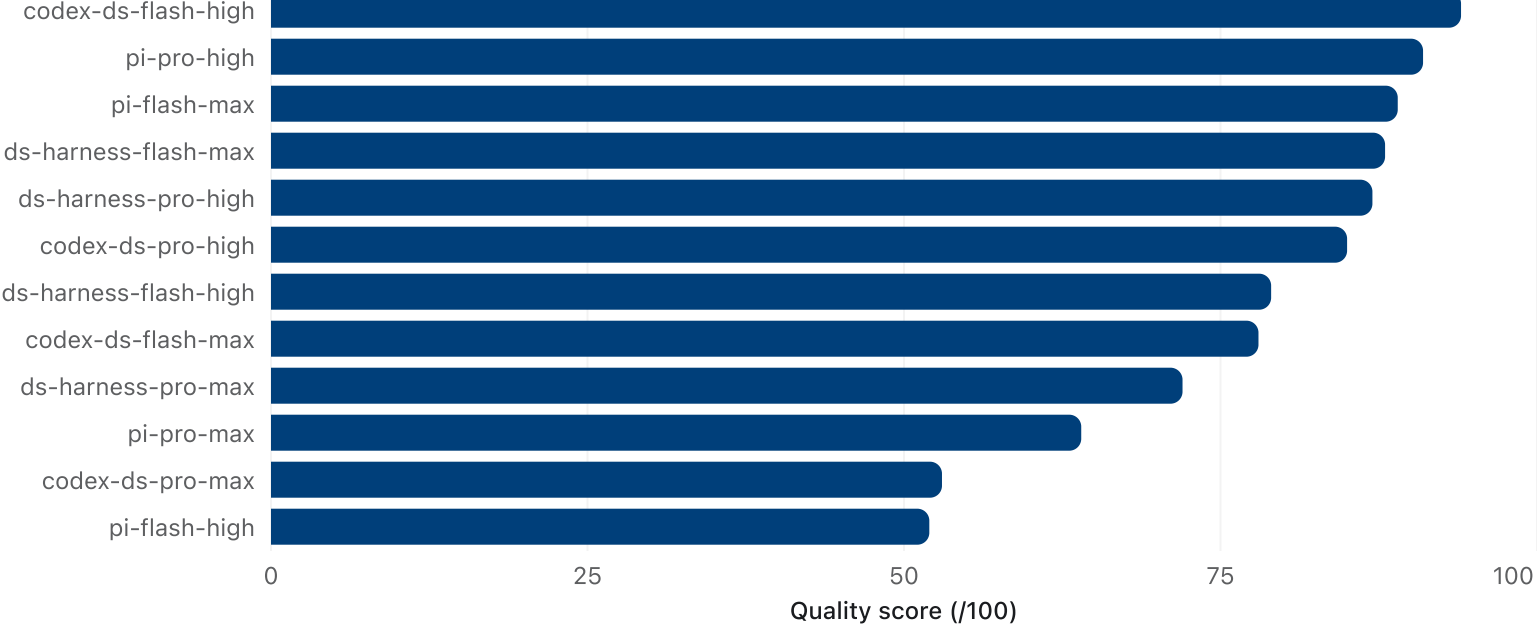
Tables: analysis/quality-assessment.md, runs/20260813193424/*/*.html, runs/20260813195515/*/*.html, runs/dsh/*/*.html

Implementation quality score

DSH Flash/max (88) and Pro/high (87) are competitive on output quality despite poor one-shot reliability.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



速度-质量前沿：DSH 当前不在默认执行前沿

吞吐首选仍是 Codex Pro/high；平衡速度与质量选 Pi Flash/max；最高实现质量选 Codex Flash/high。DSH Flash/max 和 Pro/high 的质量有竞争力，但在净时间与人工介入两个维度都被 Pi/Codex 支配，适合作为 preview CLI 的能力信号，而非默认无人值守路径。

Source: Joined performance and quality evidence

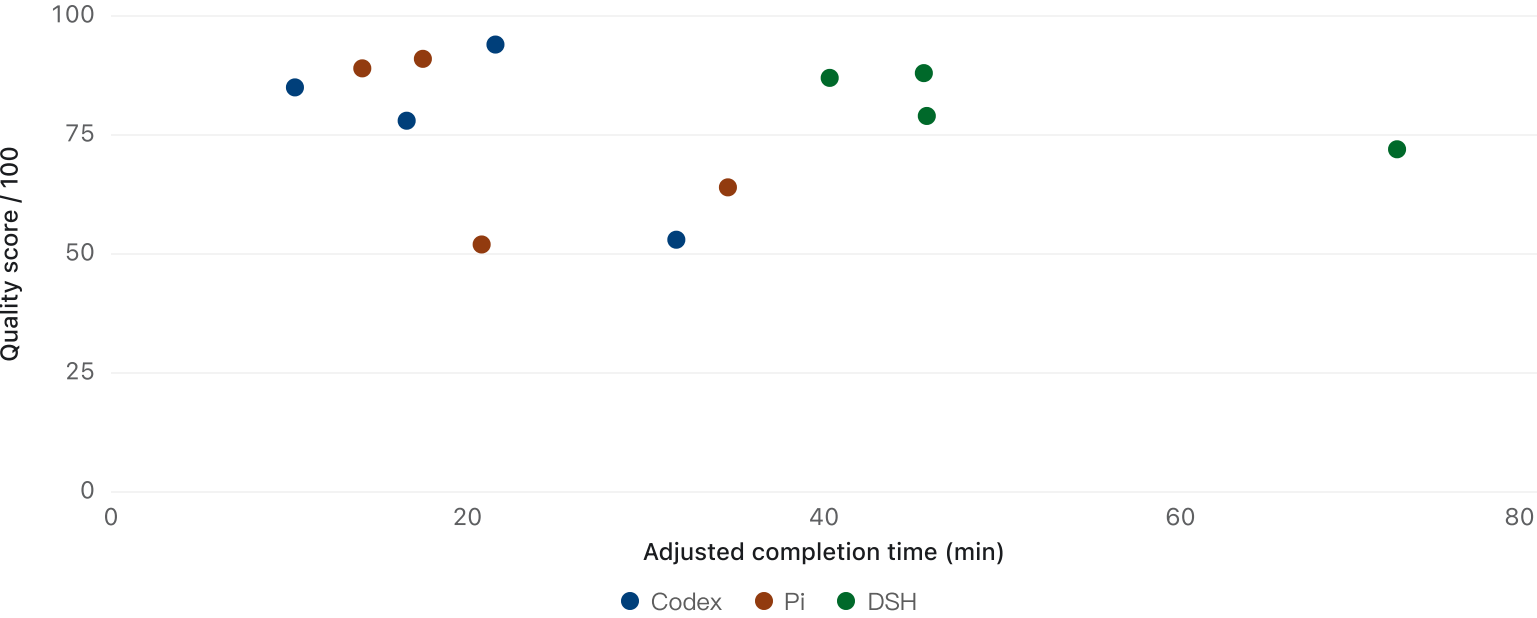
Tables: analysis/analyze-runs.mjs, analysis/quality-assessment.md, analysis/chart-map.md

Adjusted completion time vs implementation quality

DSH reaches competitive quality only after materially more active time and manual recovery.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md



Case 级证据

下表保留 adjusted/wall time、完成模式、token/cache、调用/错误/重试、质量分和浏览器发现。

Source: Joined performance and quality evidence

Tables: analysis/analyze-runs.mjs, analysis/quality-assessment.md, analysis/chart-map.md

Case-level performance and reliability

DSH adjusted time excludes only identifiable waits; wall time remains visible for audit.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md

Case	Adjusted min	Wall min	Completion	Input/cache	Model/tool calls	Tool err/calls	Retry/time out
codex-ds-pro-high	10.32	10.32	one-shot	0.76M / 97.6%	15/14	2/14	0/0
pi-flash-max	14.1	14.1	one-shot	2.62M / 99.2%	29/28	2/28	0/0
codex-ds-flash-max	16.59	16.59	one-shot	1.52M / 99.1%	23/22	7/22	0/0
pi-pro-high	17.5	17.5	one-shot	0.67M / 97.6%	8/7	1/7	0/0
pi-flash-high	20.79	20.79	one-shot	3.80M / 99.5%	41/40	6/40	0/0
codex-ds-flash-high	21.57	21.57	one-shot	2.57M / 99.2%	20/25	6/25	0/0
codex-ds-pro-max	31.72	31.72	one-shot	3.53M / 96.9%	23/32	2/32	0/0
pi-pro-max	34.61	34.61	one-shot	2.91M / 99.0%	20/19	4/19	0/0
ds-harness-pro-high	40.32	61.61	human-assisted (2 continues)	1.64M / 97.8%	15/18	0/18	3/4
ds-harness-flash-max	45.6	66.89	human-assisted (2 continues)	3.71M / 98.5%	31/34	2/34	3/4
ds-harness-flash-high	45.77	78.08	human-assisted (2 continues)	5.11M / 99.4%	49/49	4/49	4/6
ds-harness-pro-max	72.15	104.66	human-assisted (2 continues)	11.03M / 99.7%	75/77	3/77	5/6

Case-level browser quality findings

Critical status overrides the numeric quality score for deployment guidance.

Source: Reviewed case-level report query

Tables: analysis/report-source.sql, analysis/analyze-runs.mjs, analysis/quality-assessment.md

Case	Score	Status	Browser finding
codex-ds-flash-high	94	Pass	Best overall; strong exploded view, slight lower-label crowding.
pi-pro-high	91	Pass	Best composition; minor right/bottom label crowding.
pi-flash-max	89	Pass	Correct geometry; exploded top/bottom framing clips.
ds-harness-flash-max	88	Pass	Best DSH; round trip passes, minor viewport clipping.
ds-harness-pro-high	87	Pass	Clean DSH round trip; central labels overlap.
codex-ds-pro-high	85	Pass	Fastest usable; correct geometry, minor top clipping.
ds-harness-flash-high	79	Pass	Assembled shutter detached; exploded labels overlap.
codex-ds-flash-max	78	Pass	Complete; exploded framing over-zoomed and mislabelled.
ds-harness-pro-max	72	Pass	Undersized scene; labels and vertical title clipped.
pi-pro-max	64	Critical	Label fade-in throws; labels never render.
codex-ds-pro-max	53	Critical	Vertical shell caused by unrotated extrusions.
pi-flash-high	52	Critical	Vertical shell despite extensive self-tests.

Scope, definitions, and methodology

样本限定为相同 canonical prompt 的 12 次执行，每个组合仅 1 次。Pi/Codex 使用 one-shot 端到端完成时间；DSH adjusted time 从 first user 到 completed turn 的墙钟时间中，扣除日志明确记录的 300 秒 stream-idle timeout、retry backoff、以及 error turn 到下一 turn 的断线间隔。质量采用 30/30/25/15 rubric：功能、规格、视觉、验证/可维护性；关键运行时或语义几何缺陷覆盖数值排名。

Limitations and robustness

每格 n=1，不能估计方差或显著性。DSH 部分 turn 缺少 end 事件，adjusted time 只能扣除可明确识别的等待，仍可能包含残余等待或重叠执行。Pi/Codex 日志没有 DSH 同口径的 LLM retry 事件，因此 retry 数只用于 DSH 内部诊断。工具错误率按显式 error result 计数；它衡量发生率，不衡量严重程度。质量含人工视觉判断，DSH 主模型不具备图像输入能力。

Recommended operating policy

1. 当前默认仍使用 Codex + DeepSeek V4 Pro + high。
2. DSH 在无人值守 benchmark 中先设 one-shot completion gate；P0 修复前不以最终 quality 分掩盖续跑失败。
3. 禁止自动升级到 Pro/max：三个 harness 中都更慢，且没有稳定质量收益。
4. 下一轮每格至少 5 次，记录 p50/p90、one-shot rate、automatic recovery rate、TTFT、成本与失败 taxonomy。